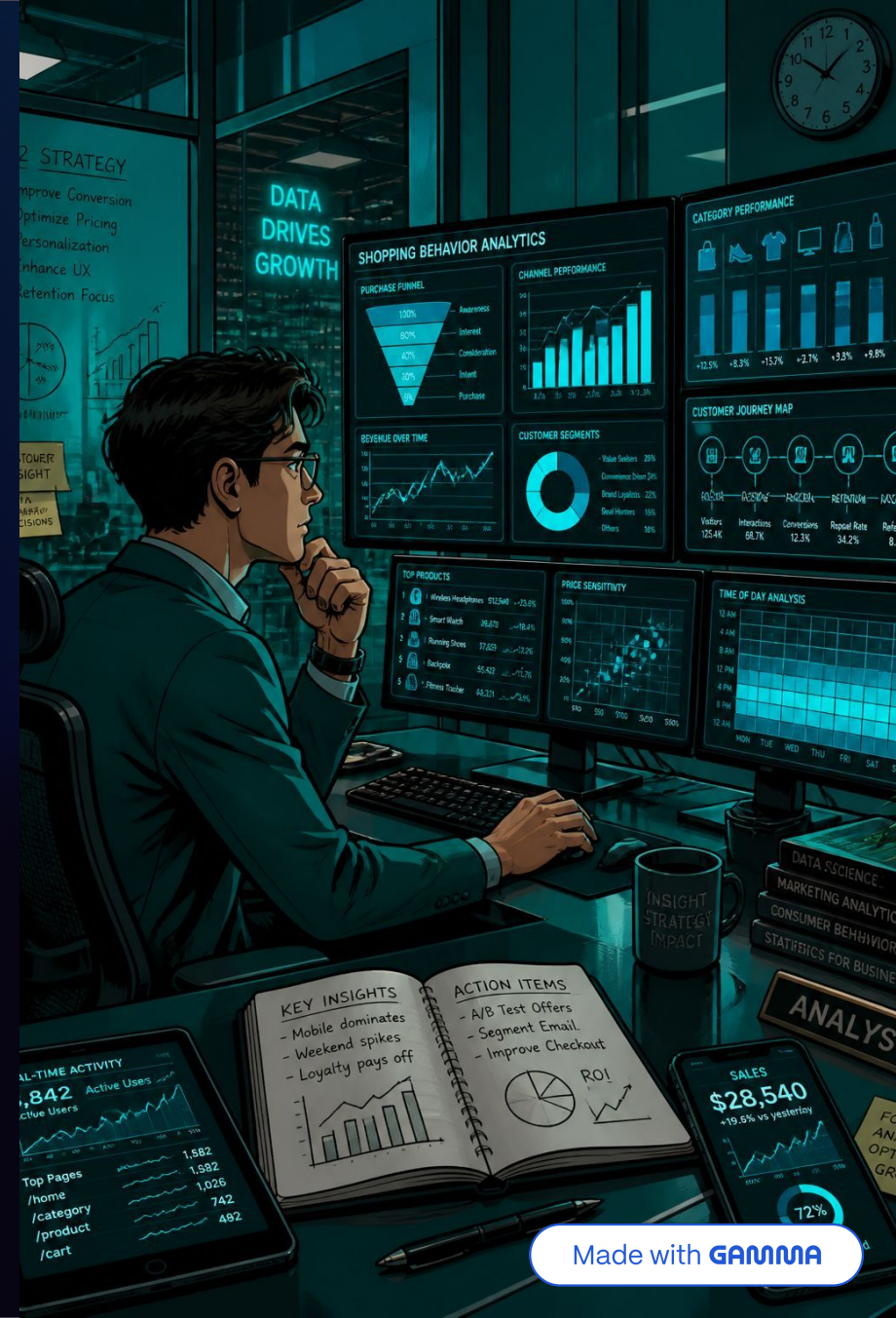


Customer Shopping Behavior Analysis

Analyzing **3,900 transactions** across product categories to uncover spending patterns, customer segments, and subscription behavior — guiding strategic business decisions.



Dataset at a Glance

3,900

Total Purchases

Transactional records analyzed

18

Data Columns

Demographics, purchases & behavior

37

Missing Values

Review Rating column only

\$59.76

Avg. Purchase

Average transaction value (USD)

Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season), and shopping behavior (Discount Applied, Review Rating, Shipping Type, Frequency).

Python: Data Preparation & Cleaning

Exploration

Loaded via `pandas`; used `df.info()` and `.describe()` for structure and summary stats.

Missing Data

37 nulls in Review Rating imputed using the **median rating per product category**.

Feature Engineering

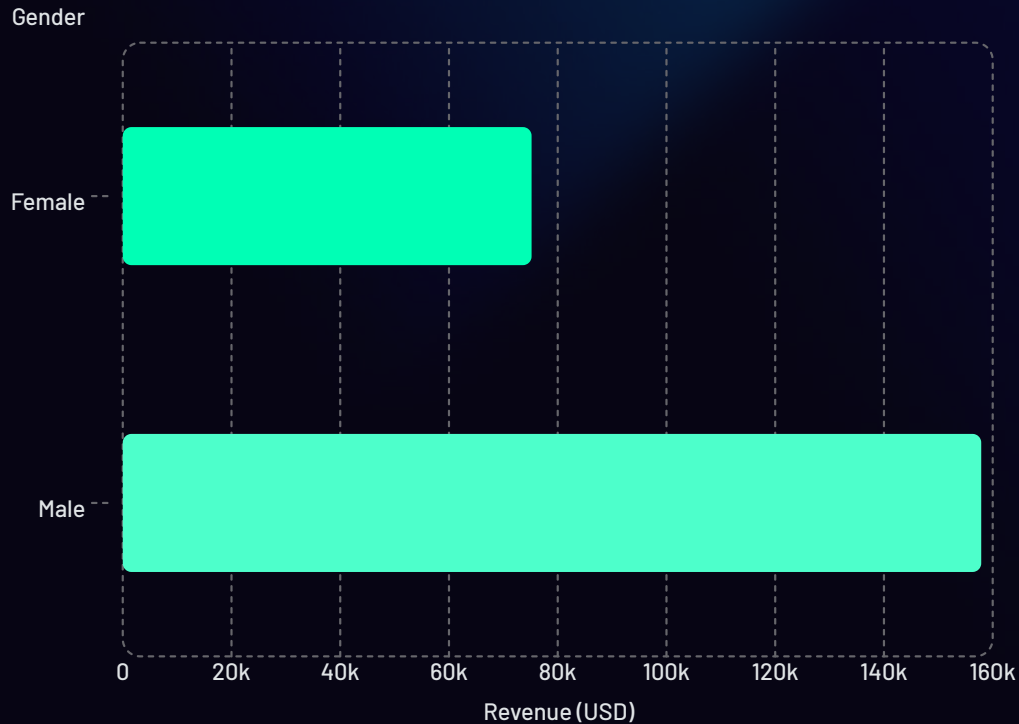
Created `age_group` (binned ages) and `purchase_frequency_days`. Dropped redundant `promo_code_used`.

Database Integration

Cleaned DataFrame loaded into **PostgreSQL** for SQL-based analysis.



SQL Analysis: Revenue & Spending



Revenue by Gender

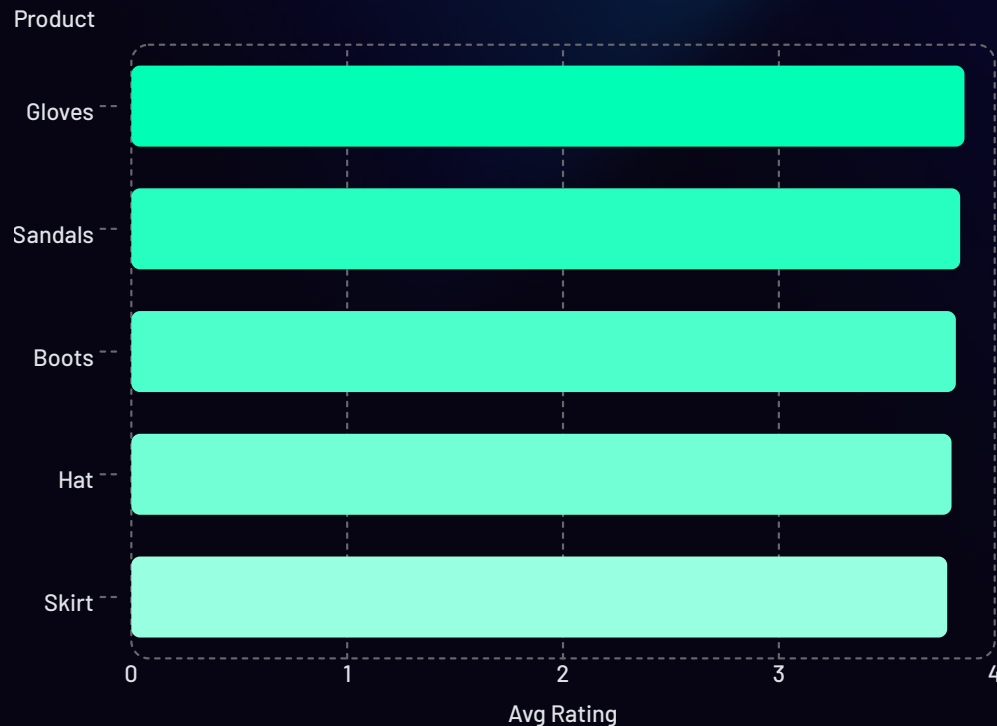
Male customers generated **\$157,890** in revenue — more than double female customers at \$75,191.

High-Spending Discount Users

839 customers used discounts yet spent above the average purchase amount — indicating price-sensitive but high-value buyers.

SQL Analysis: Products & Shipping

Top 5 Products by Rating



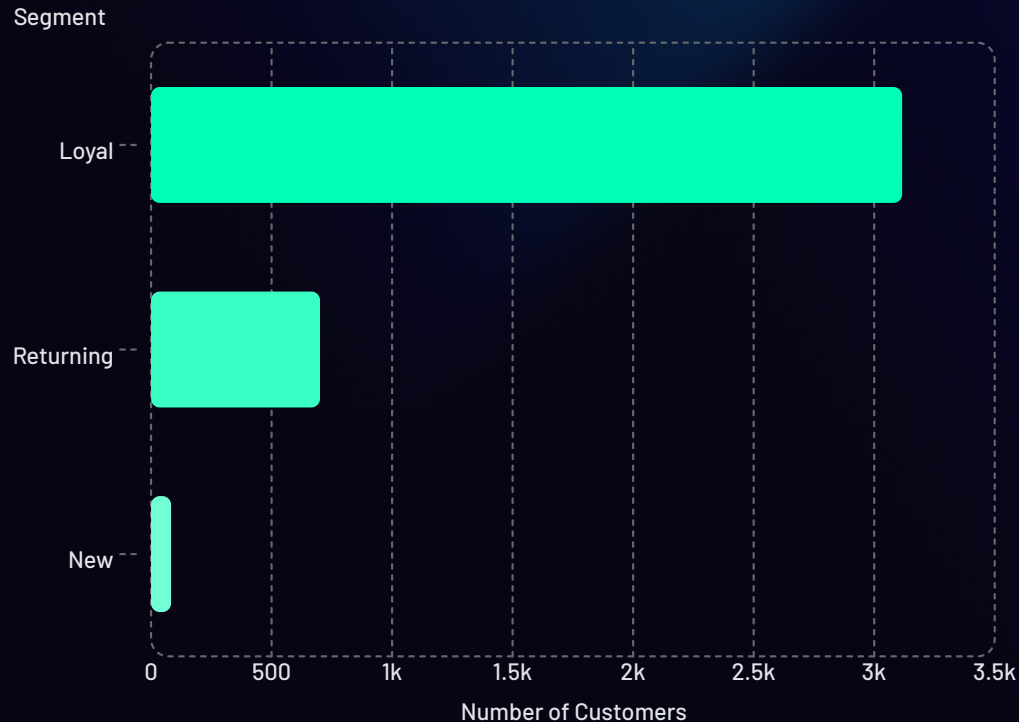
Shipping Type Comparison

Express shipping averaged **\$60.48** vs. Standard at \$58.46 — a modest but consistent premium.

Discount-Dependent Products

Hats (50%), Sneakers (49.66%), and Coats (49.07%) had the highest discount rates — suggesting strong price sensitivity in these categories.

Customer Segmentation



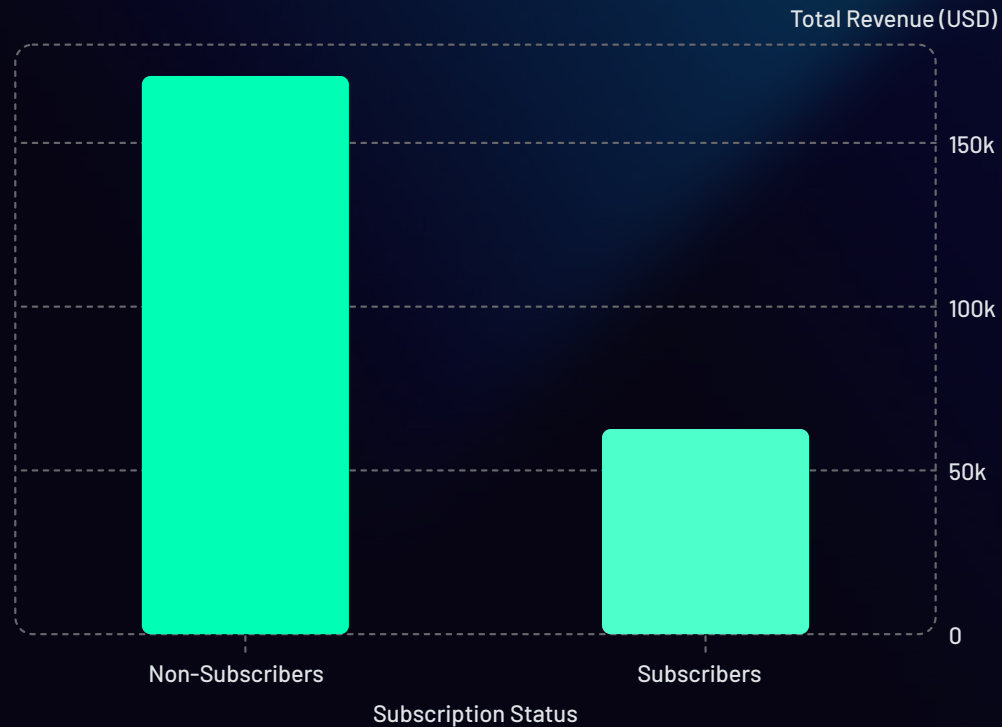
Segment Breakdown

Customers classified as **New**, **Returning**, and **Loyal** based on purchase history.

- **Loyal: 3,116** — the dominant segment, representing the core revenue base
- **Returning: 701** — active but not yet fully loyal
- **New: 83** — smallest segment, key growth opportunity

Of customers with **>5 purchases**, 958 are subscribers vs. 2,518 non-subscribers — repeat buyers show meaningful subscription uptake.

Subscription Status & Revenue



Subscribers vs. Non-Subscribers

Only **27%** of customers are subscribers, yet they represent a high-value segment worth growing.

Non-Subscribers

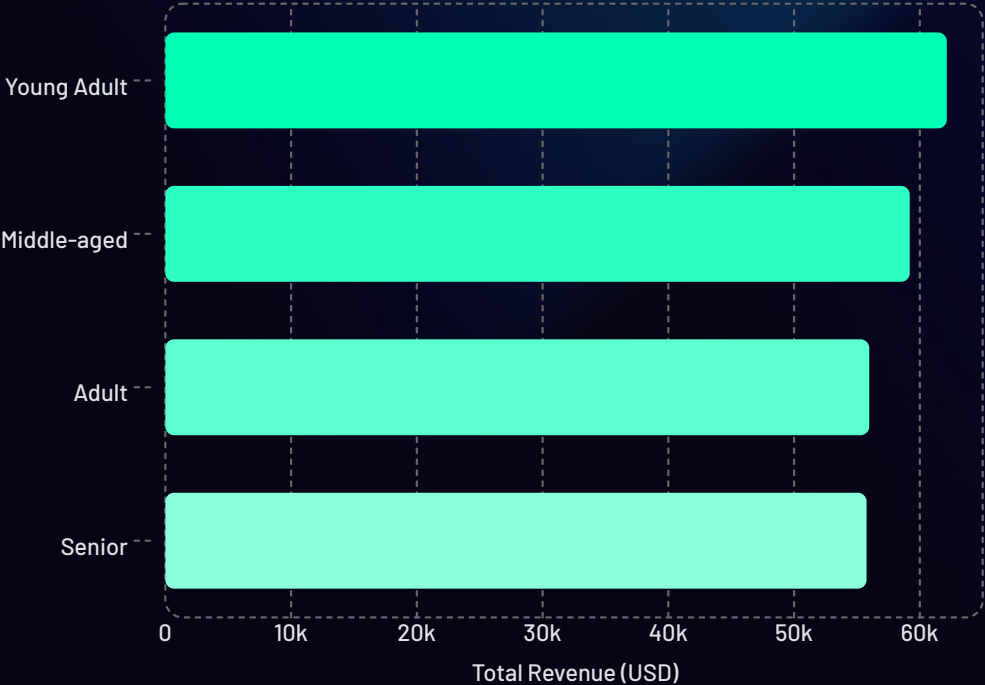
2,847 customers · \$170,436 revenue

Subscribers

1,053 customers · \$62,645 revenue

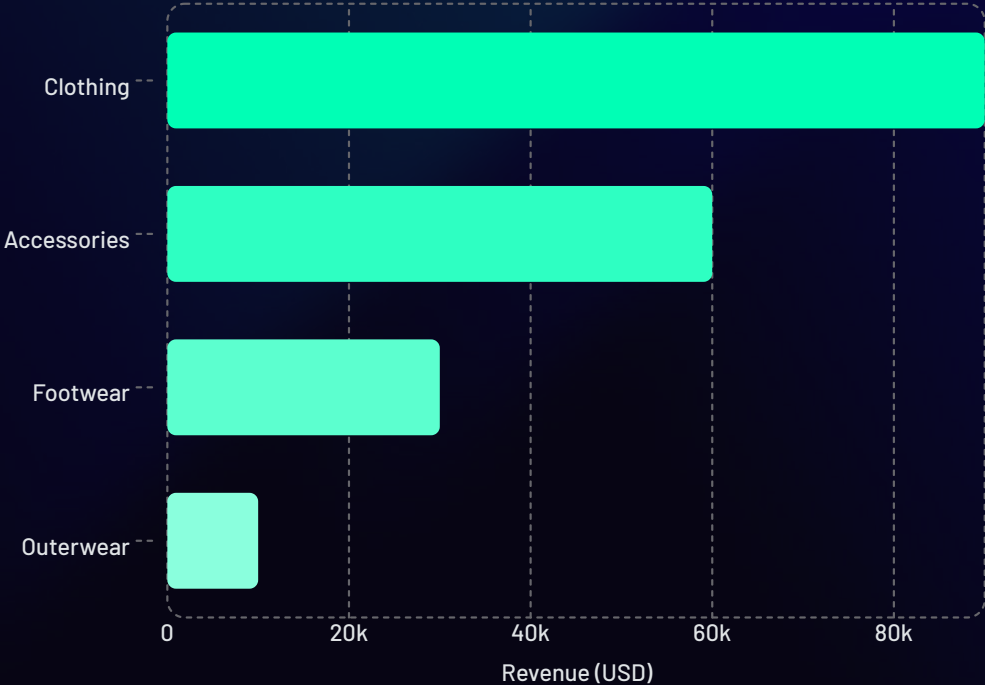
Revenue by Age Group & Category

Age Group



Revenue by Category

Category



Young Adults lead revenue contribution. **Clothing** dominates all categories at ~\$90K.

Power BI Dashboard



Customer Behavior Dashboard

An interactive Power BI dashboard presents all key insights visually with filters for subscription status, gender, category, and shipping type.

Visualizations include subscription breakdown (27% Yes / 73% No), revenue and sales by category, and revenue and sales by age group.

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to grow the 27% subscriber base.



Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.



Review Discount Policy

Balance sales boosts with margin control, especially for discount-dependent products.



Targeted Marketing

Focus on high-revenue age groups and express-shipping users; highlight top-rated products in campaigns.

